

Discover U: The Impact of Gamified, AI-Supported Career Counseling on Student Career

Alignment

Comparing Experiential Guidance Platforms with Traditional Counseling Methods

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Section 1: Context

Every year, thousands of students graduate from high school or college still unsure of what comes next. A 2017 Gallup-Strada survey reported that **36% of U.S. college graduates regret their choice of major**, and more than half admitted they would make different decisions if given the chance (Gallup-Strada, 2017). Students often grapple with their actual interests or passions, and what will net them a well-paid career; we've all heard of students who wish they would have majored in something that was more marketable to employers. This persistent mismatch between academic choices and long-term goals suggests that our current systems of career guidance are often falling short.

Anecdotally, among my own peers in high school, I have had friends express interest in a certain passion, only to ask the question, "Can I make a career out of this?" Counselors are often well-informed on popular career paths, such as law or medicine, but may have little input on burgeoning careers or industries; in these cases, mid-career school counselors may be of little help; and may be overburdened with students to do a deep dive on unfamiliar career paths.

One major issue is the way counseling is structured in schools. High school counselors, for example, typically manage caseloads of nearly 500 students each (NACAC, 2016). Under those conditions, individualized exploration is limited. Guidance tends to come in the form of quick interest surveys or brief personality assessments, which may be useful at a surface level but rarely provide students with a sense of what a career actually looks and feels like (Gottfredson, 2005; Holland, 1997). These quick interest surveys pretty much ask a student what they may be interested in, but they're not very personal, and this must especially be true at public schools where the caseload for the counselors is higher.

This is where gamification becomes relevant. Studies show that when information is presented with rewards, levels, or progress markers, students tend to persist longer and engage more meaningfully (Dodson, 2018; Smith & Allen, 2020). Kenneth Dodson, in particular, found that gamified counseling can push students to explore options with greater curiosity and resilience.

Other countries are already experimenting in this space. In the United Kingdom, the Forage platform lets students test their skills through simulations built by companies. These experiences do not simply “tell” a student they might fit into consulting or finance, but they let them try it in a low-stakes environment (Forage, 2021). Intuitively, this makes a lot of sense, as students actually get to try out their future career before committing to it.

The pressure on American students is well documented. Jessica Byrant (2019) notes that **over half of high school students feel intense pressure** to make long-term decisions during adolescence, often without strong support. If the stakes are that high, then tools which are interactive, personalized, and even enjoyable may lead to better outcomes than the traditional quick counseling appointment.

This study investigates whether a gamified, AI-supported counseling model can offer a clearer, more engaging sense of career alignment than conventional approaches.

Section 2: Research Question

Can the gamification of job counseling provide more accurate and engaging assessments of educational and career paths than traditional counseling, particularly for students who feel uncertain or unsatisfied with their options?

Section 3: Theory

Holland's Theory of Vocational Personalities (RIASEC) remains a cornerstone of career counseling. It organizes individuals into six broad categories: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional. Each group points to particular work environments. Realistic types thrive in hands-on fields like engineering or construction, while Investigative types are drawn toward research and scientific inquiry (Holland, 1997).

Its appeal is obvious: it is straightforward, intuitive, and widely used. Yet its limits are equally obvious. Gottfredson's theory of Circumscription and Compromise, for instance, highlights how social expectations and external constraints shape what options people actually pursue (Gottfredson, 2005). In practice, both theories provide broad categories but stop short of offering a dynamic process of trial, error, and adjustment.

That is exactly the gap Discover U seeks to address. By merging Holland's framework with interactive simulations, the platform offers a hybrid: a personality-driven starting point combined with experiential exploration. Instead of locking students into a "type," Discover U creates space for adaptability, iteration, and feedback, the qualities traditional counseling models often lack.

Section 4: Methodology

Participants

This study involved **240 high school juniors** across three public schools. Students were randomly divided into two groups:

- **Gamified Group (n = 120):** Used Discover U for four weeks, engaging with simulations, badges, and progress tracking.
- **Traditional Group (n = 120):** Completed a single 45-minute counseling session using a Holland-based interest inventory.

The sample included 52% female, 48% male, and 38% first-generation college students.

Procedure

Both groups first completed a **baseline survey** on career confidence, clarity, and preparedness.

Afterward:

- Gamified group → four weeks with Discover U.
- Traditional group → one-time counseling report.

All participants then completed a **post-survey** and a follow-up four weeks later.

Measures

- Career satisfaction and understanding
- Confidence in decision-making
- Preparedness and path clarity
- Persistence of interest

- Qualitative reflections (open-ended responses, focus groups)
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Section 5: Results

Career Satisfaction & Understanding

Before the intervention, only about **41% of students said they were satisfied** with prior counseling. Post-survey data showed that Discover U participants reported significantly higher satisfaction, often describing the experience as “*interactive*,” “*motivating*,” or “*a way to see real pathways*.” By contrast, traditional counseling students tended to remain neutral or dissatisfied, with some calling the session “*too short*” or “*generic*.”

Confidence in Decision-Making

At baseline, students across both groups expressed modest confidence, often unsure if they could make informed choices. After four weeks:

- **Gamified group:** Students frequently wrote that the platform “*boosted trust in my abilities*” and “*made me less anxious about the future*.”
- **Traditional group:** Confidence levels barely shifted, and reflections suggested little additional clarity.

Career Exploration & Persistence

Students in the gamified group often discovered new interests:

- *“Opened my mind to careers I never considered.”*
- *“I discovered I like teamwork more than I thought.”*

Traditional group responses were flatter, usually describing the session as surface-level and limited.

Preparedness & Path Clarity

Gamified students repeatedly highlighted the **hands-on nature** of the tool:

- *“It provided actual pathways and mentors.”*
- *“Hands-on practice was missing before.”*

Traditional group students expressed frustration that their counseling experience ended without actionable next steps.

Subgroup Trends

- **First-generation students** in the gamified group showed notable gains in self-efficacy.

- **Female students** reported increased openness to STEM fields, echoing prior research on gamification's potential to reduce barriers.
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Section 6: Analysis

Taken together, the results suggest Discover U offered a **qualitatively different kind of engagement** than traditional counseling.

1. **Career Alignment:** Rather than reinforcing existing preferences, the gamified platform pushed students to explore and refine new, personally relevant options.
2. **Engagement:** Usage logs and reflections showed greater depth and time commitment in the gamified group. Traditional counseling, by contrast, often felt like a one-time requirement.
3. **Confidence:** Post-survey reflections show that gamification improved self-confidence in career decision-making. This shift is striking given how low baseline confidence was across both groups.
4. **Persistence:** Students from the gamified group continued thinking about and engaging with their chosen paths weeks later — a sign of longer-term impact.
5. **Equity Impact:** First-generation and female students benefited disproportionately, highlighting the platform's potential to narrow existing gaps in career guidance.

Section 7: Discussion

Implications

If scaled, gamified platforms like Discover U could:

- **Increase Engagement:** Move beyond one-off sessions into ongoing exploration.
- **Promote Equity:** Offer individualized support even in schools where counselor ratios approach 500:1 (NACAC, 2016).
- **Bridge Theory and Practice:** Link personality models with simulations that let students “try on” careers.
- **Boost Long-Term Satisfaction:** Reduce the widespread regret many graduates express about academic and career choices.

Limitations

Naturally, no study is without its caveats.

- Simulations cannot fully capture the complexity of real jobs.
- Unequal technology access could reinforce disparities.
- Privacy and data handling remain important concerns.
- Some students may not be motivated by gamification elements.

Rebuttals

Still, these challenges are not insurmountable.

- The platform is meant to **complement, not replace** human counselors.
 - Partnerships and grants can expand access.
 - Data protections can be embedded from the start.
 - Multiple motivational features can ensure different learning styles are accommodated.
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Section 8: Conclusion

This study asked whether gamified counseling could outperform traditional methods in accuracy, engagement, and confidence. The answer appears to be yes. Students using Discover U showed **higher satisfaction, stronger confidence, and sustained interest** in career exploration compared to their peers in traditional sessions. This makes a lot of sense since counseling appointments are cursory and superficial, asking students what they think may sound interesting, through a paper form or quick meeting, but they lack real engagement with the student. As mentioned earlier from my own anecdotal experience in high school, a lot of students may have interests that go beyond the scope of popular, mainstream careers; and this is where counselors

are especially ill-prepared for helping students explore. Often, it is not their fault, but simply that they have many other students they must look after.

When students rely on these traditional counseling methods, they often don't get more exposure to a career other than its title: for example, a short form may have a job title listed called "finance professional" – begging the question of: what kind of finance professional? Investment banker? Trader? Venture capitalist? Real estate private equity? The counselors themselves often don't know what half of these careers are, especially in low-income areas. Thus, a really key part of this gamified program is getting the chance to dive deep; relying less on students' and counselors' preconceived notions of what these careers may entail.

By blending Holland's framework with gamified simulations, Discover U offers a scalable and engaging alternative to the static models that dominate most schools. Its promise is especially strong for first-generation students and underrepresented groups, where access to effective guidance is often lacking. It is my dream to expand this program to more of these types of students, who will benefit most from more guidance.

There are challenges to consider, from equity of access to the realism of simulations, but the evidence suggests the benefits outweigh the drawbacks. If adopted widely, Discover U could reshape how students navigate one of the most important choices of their lives: aligning their strengths, interests, and opportunities with a sustainable career path.

References

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